



AALBORG UNIVERSITY  
DENMARK

Written examination in the course

Engineering Optimisation - Concepts, Methods and Applications

Monday June 8th 2015

kl. 9.00 - 13.00 (4 hours)

All usual helping aids are allowed, i.e. books, notes, calculator, computer etc. All communication equipment and computer communication protocols must be turned off.

You may answer the questions in English or Danish.

**REMEMBER** to write your study number and page number on all sheets handed in.

The set consists of 7 exercises. The total weighting for each of the exercises is stated in percentage. You need 50 % in order to pass the exam.

*It should be clear from the solution, which methods are used, and there should be a sufficient number of intermediate calculations, so the line of thought is clear.*

### Exercise 1: (10 %)

A company wants to maximize its day-to-day profit from manufacturing electrical motors. Formulate the profit maximization problem as a standard optimization problem based on the following information:

- The company produces two motors: 1) Type A and 2) Type B.
- To maintain market shares the company needs to produce at least 30 motors of Type A every day and at least 30 motors of Type B every day.
- The daily cost of producing Type A is a fixed cost of 1000 Euro and a further 150 Euro per motor. The daily cost of producing Type B is a fixed cost of 1500 Euro and a further 225 Euro per motor.
- The company has a total of 5000 man hours available per day. It takes 20 man hours to produce a motor of Type A and 24 man hours to produce a motor of Type B.
- The company sells the motors at the following prices: 300 Euro per motor of type A and 400 Euro per motor of type B.

Note: you should only formulate the problem, not solve it.

### Exercise 2: (15 %)

We will consider gradient-based minimisation of the following unconstrained function:

$$f(\mathbf{x}) = 2(x_1^2 - x_2) + x_1^2 - 2x_1 + 5 \quad (1)$$

The starting point is:  $\mathbf{x}^{(0)} = [-1 \ 3]^T$ .

a) Complete the first iteration of the steepest descent method for the function. The 1D line search problem should be solved analytically.

b) Can Newton's method be applied for determining the search direction in iteration 1?

If yes, then determine the search direction.

If no, then state an alternative robust method for determining the search direction.

### Exercise 3: (9 %)

An optimisation problem is formulated as:

$$\text{minimise} \quad f(\mathbf{x}) = 2x_1^2 + 6x_2^2 - 3x_1x_2 - x_1 + 2x_2$$

Subject to:

$$\begin{aligned} g_1(\mathbf{x}) &= x_1 - 12 && \leq 0 \\ g_2(\mathbf{x}) &= 7x_1 - 2x_2 - 4 && \leq 0 \\ g_3(\mathbf{x}) &= x_1 + x_2 - 20 && \leq 0 \end{aligned}$$

The initial starting point for an optimisation is  $\mathbf{x}^{(0)} = (0.5, 0)$ , for which a search direction is determined to  $\mathbf{d}^{(0)} = [-1 \ -0.5]^T$  and the vector of Lagrange multipliers for the constraints is  $\mathbf{u} = [0 \ 0 \ 0]^T$ . Let  $R_0 = 1$  and  $\gamma = 0.5$ . Choose the trial step according to the sequence  $t_0 = 1$ ,  $t_1 = \frac{1}{2}$ ,  $t_2 = \frac{1}{4}$ ,  $t_3 = \frac{1}{8} \dots$

Calculate the step size using the inexact line search procedure (approximate step size procedure) and determine the new design variables,  $\mathbf{x}^{(1)}$ , for the next iteration.

#### Exercise 4: (9 %)

Solve the following problem by setting up the solution tree and using the *Local Minimization Branch & Bound Method*:

$$\text{minimise} \quad f(\mathbf{x}) = -6x_1 - 3x_2$$

Subject to:

$$g_1(\mathbf{x}) = 0.5x_1 + x_2 - 11 \leq 0$$

$$g_2(\mathbf{x}) = 2.5x_1 + x_2 - 20 \leq 0$$

$$g_3(\mathbf{x}) = 2x_1 - x_2 - 10 \leq 0$$

Both  $x_1$  and  $x_2$  should be integer values

As a help for solving the problem the objective function contours and constraints are plotted in figure 1.

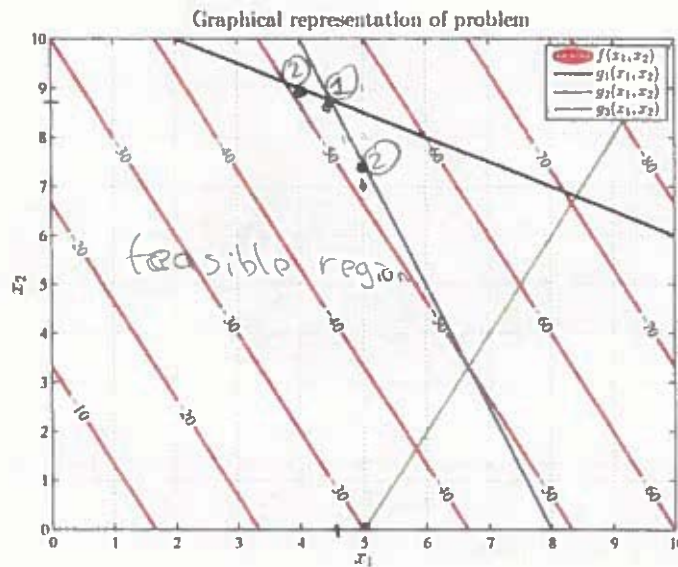


Figure 1: Graphical representation of problem in exercise 4.

#### Exercise 5: (7 %)

Solve the following linear optimisation problem using the basic steps of the Simplex method and tableau's:

$$\text{minimise} \quad f(\mathbf{x}) = 3x_1 - 2x_2 \quad (2)$$

Subject to the constraints:

$$\begin{aligned} g_1(\mathbf{x}) &= -2x_1 + x_2 \leq 5 \\ g_2(\mathbf{x}) &= x_1 + 4x_2 \leq 3 \\ x_1 &\geq 0 \quad \forall \quad x_1 = \{1, 2\} \end{aligned} \quad (3)$$

### Exercise 6: (20 %)

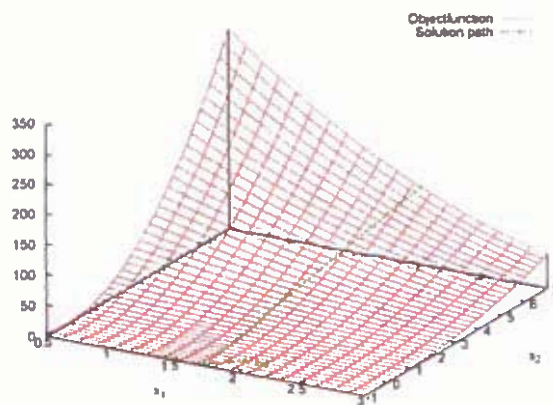
The cost function  $f(\mathbf{x})$  is defined as the sum of squares of the residual function  $\mathbf{r}(\mathbf{x})$

$$f(\mathbf{x}) = \frac{1}{2} \sum_{i=1}^m r_i^2 \quad (4)$$

where

$$\mathbf{r}(\mathbf{x}) = \begin{bmatrix} x_1 - 2 \\ (x_1 - 2\beta)x_2 \\ x_2 + 1 \end{bmatrix} \quad (5)$$

The initial cost function  $f(\mathbf{x}_0) = 130$  for  $\mathbf{x}_0 = [2, 7]$  and the constant  $\beta = 2$ . The minimum value  $f(\mathbf{x}^*) \approx 0.396478$  at  $\mathbf{x}^* \approx [2.0882, -0.2148]$ , see figure 2.



Solutions  
Lecture 10

Figure 2: Surface plot of the cost function and the trajectory produced by an iterative solution scheme.

The following questions must be answered:

1. Derive an analytical expression for Jacobian matrix associated with equation 5.
2. Numerical solutions are expected for the following two sub-questions, where the initial parameters  $\mathbf{x}_0 = [2, 7]$  are inserted into the expressions derived above.
  - (a) Using the initial parameters  $\mathbf{x}_0 = [2, 7]$ ; calculate the gradient and approximate the Hessian using only first order information from the Jacobian matrix derived above in question 1.
  - (b) Complete the first 3 iterations using the Levenberg-Marquardt algorithm (where the step was defined as  $\mathbf{s}_k = -(\mathbf{H} + \lambda \mathbf{I})^{-1} \nabla f(\mathbf{x}_k)$ ). Start the iterations using an initial damping parameter  $\lambda = 10$  and report  $\lambda$ , object-function  $f(\mathbf{x}_k)$  and  $\mathbf{x}_k$  for the first 3 iterations.

| iter. | $\mathbf{x}$ | $f(\mathbf{x})$ | $\lambda$ |
|-------|--------------|-----------------|-----------|
| 0     | $[2, 7]$     | 130             | 10        |
| 1     |              |                 |           |
| 2     |              |                 |           |
| 3     |              |                 |           |

possible to add  $\nabla$   
 $\rightarrow$   $\nabla$  og  $J$  defined in ParaExamGrad1  
 run GradHess.m  
 $\rightarrow$   $\mathbf{r}$  and  $J$  defined in Exam2015  
 run SolveExam1.m  
 Also have Plot.m and Taylor.m there.

DOF

$$3 \cdot n - 2 \cdot l$$

$$3 \cdot 2 - 2 \cdot 2 = 2$$

### Exercise 7: (15 %)

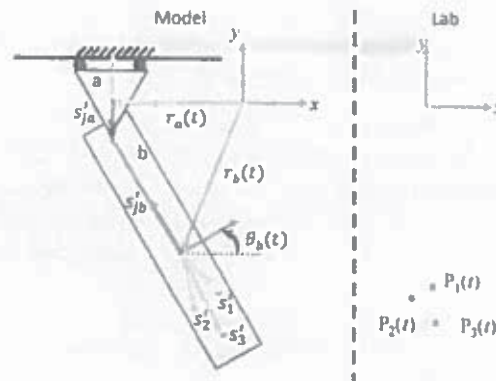


Figure 3: A mechanical system consisting of two rigid bodies.

Figure 3 shows two rigid bodies (a and b) in a two-dimensional space that are connected to each other with a revolute joint and body a is connected to ground with a translational joint. Three markers (the green dots in the figure) are attached to body b with local coordinates,  $s'_1$ ,  $s'_2$  and  $s'_3$  given in the indicated, body-fixed reference frame of body b (indicated in red in the figure). The location of the revolute joint in body-fixed coordinates of bodies a and b are given by  $s'_{ja}$  and  $s'_{jb}$ , respectively. The global reference frame is indicated in blue.

The position and orientation of the bodies are to be described in terms of a full cartesian formulation, i.e. position and orientation of the bodies relative to the global reference frame. The global position and rotation angles of the bodies are denoted,  $r_a(t) = \begin{bmatrix} x_a(t) \\ y_a(t) \end{bmatrix}$ ,  $r_b(t) = \begin{bmatrix} x_b(t) \\ y_b(t) \end{bmatrix}$ ,  $\theta_a(t)$  and  $\theta_b(t)$  as indicated on the figure. Please note that  $\theta_a(t)$  is not indicated in the figure since it is assumed to always remain zero but a constraint equation to specify this must be included.

A motion capture experiment was performed, where the positions of the markers were measured in the global reference frame for a duration of time and the measured coordinates are denoted  $P_1(t)$ ,  $P_2(t)$  and  $P_3(t)$ .

From the measured trajectories of the markers, it is desired to set up a nonlinear least-squares optimisation problem that can be used to compute the position and orientation of the rigid bodies. That is, the least-square difference between the measured markers and the points on the bodies is minimised while ensuring that the revolute and translational joint constraints are fulfilled for each time step in the analysis.

The following questions must be answered:

1. Is the system of kinematic equations under-determinate, determinate or over-determinate when the motion of the system is driven by the measured markers? Justify the answer by counting the number of degrees-of-freedom of the mechanical system compared to the number of kinematic driver equations.
2. Formulate a constrained nonlinear least-squares optimisation problem that can be used to compute both the positions,  $r_a(t)$  and  $r_b(t)$ , and rotations,  $\theta_a(t)$  and  $\theta_b(t)$ , simultaneously.
3. Derive the first-order optimality conditions for the optimisation problem derived in question 2.

### Exercise 8: (15 %)

We assume you are going to solve a design optimisation problem where the maximum pressure of fluid running in a network of pipes is to be minimized.

The pipe system consists of  $n = 10$  pipes, the design variables of the problem are denoted  $x_i$ ,  $i = 1, \dots, n$ , and these are the cross sectional areas of the pipes with prescribed minimum values  $\underline{x}_i$ ,  $i = 1, \dots, n$ , and maximum values  $\bar{x}_i$ ,  $i = 1, \dots, n$ . We assume that you have been given an analysis model that can compute the pressure distribution in the network of pipes, and this will give you pressure values  $P_i$ ,  $i = 1, \dots, n$ , related to the pipes. Gradients can also be computed by the analysis code.

- a) Explain the main mathematical difficulty of solving the optimization problem of minimizing the maximum pressure in the  $n$  pipes.
- b) State the mathematical formulation for three fundamentally different ways of formulating the optimization problem. Describe advantages and disadvantages of the three chosen optimisation problem formulations.



## Exercise 1 (10%)

A company wants to maximize its day-to-day profit from manufacturing electrical motors. Formulate the profit maximization problem as a standard optimization problem based on the following information

Info:

- The company produces two motors: 1) Type A and 2) Type B
- To maintain market shares the company needs to produce at least 30 motors of Type A every day and at least 30 motors of Type B every day
- The daily cost of producing type A is fixed cost of 1000 euro and a further 150 euro per motor. The daily cost of producing Type B is a fixed cost of 1500 euro and a further 225 euro per motor
- The company has a total of 5000 man hours available per day. It takes 20 man hours to produce a motor of type A and 24 man hours to produce motor of type B.
- The company sells the motors at: 300 for A  
400 for B

Define Design variables:

$n_A$ : Number of A motors  
 $n_B$ : Number of B motors

Standard optimization Design model

$$\text{minimize: } f(n_A, n_B) = \frac{1}{-1000 - 1500 + (300 - 150) \cdot n_A + (400 - 225) \cdot n_B}$$

Subjected to:

$$n_A \geq 30$$

$$30 - n_A \leq 0$$

$$n_B \geq 30$$

$$30 - n_B \leq 0$$

$$20 \cdot n_A + 24 \cdot n_B \leq 5000 \Rightarrow 20 \cdot n_A + 24 \cdot n_B - 5000 \leq 0$$

## Exercise 2

We will consider gradient-based minimisation of the following unconstrained function.

$$f(x) = 2(x_1^2 - x_2) + x_1^2 - 2x_1 + 5$$

The starting point is:  $x^{(0)} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}^T$

a) complete the first iteration of the steepest descent method for the function. If line search problem should be solved analytically.

Step 1  $x^{(0)} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$   $b=0$   $\epsilon = 0.001$   $f(x^{(0)}) = 4$

Step 2 calculate the gradient at the starting point

$$c(x) = \begin{bmatrix} 6x_1 - 2 \\ -2 \end{bmatrix} \quad c^{(0)} = \begin{bmatrix} -8 \\ -2 \end{bmatrix}$$

Step 3 The norm of the gradient

$$\|c^{(0)}\| = \sqrt{(-8)^2 + (-2)^2} = 2\sqrt{17} > \epsilon \Rightarrow \text{starting point is not a minimum}$$

Step 4 From the gradient the direction of steepest descent is found

$$d^{(0)} = -c(x^{(0)}) = -c^{(0)} = \begin{bmatrix} 8 \\ 2 \end{bmatrix}$$

Step 5 Next the step size is computed. Analytically the problem is transformed into a 1D line search problem

$$f(x) = f(x^{(0)} + \alpha d^{(0)}) = 2 \cdot (-1 + 8\alpha)^2 - 2 \cdot (3 + 2\alpha) + 5 = 2\alpha^2 + 4$$

$$c = \begin{bmatrix} 16\alpha \\ 2 \end{bmatrix} \quad d^{(0)} = \begin{bmatrix} 8\alpha \\ \alpha \end{bmatrix} \quad = 16\alpha^2 - \alpha + 4$$

then we use it to obtain  $\alpha$  (line search)

$$c = 3 \quad 16\alpha - 1 = 0 \Rightarrow \alpha = \frac{1}{16} \approx 0.0625$$

Step

the new direction

$$x^{(1)} = x^{(0)} + \alpha d^{(0)} = \begin{bmatrix} -1 \\ 3 \end{bmatrix} + \frac{1}{16} \begin{bmatrix} 8 \\ 2 \end{bmatrix}$$

$f(x^{(1)}) =$



Can Newton's method be applied for determining the search

direction in iteration 1?

If yes, then determine the search direction

If no, then state an alternative robust method for determining the search direction.

Step 3 is the first step here (Lecture 4 slide 15)

Calculate the Hessian

$$H = \begin{bmatrix} 6 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{this matrix is singular} \\ \Rightarrow \text{it is not possible to determine a search direction (slide 16 - Lecture 4)}$$

Alternative methods

Use the conjugate gradient method

$$d^{(k)} = -\nabla f(x^{(k)}) + \beta_k d^{(k-1)}$$

where

$$\beta_k = \left( \frac{\|c^{(k)}\|}{\|c^{(k-1)}\|} \right)^2$$

Uses only gradient info and should always be used rather than the steepest descent method.

(see slides 8-13 Lecture 4) for the method and everything)

## Exercise 13.1

An optimisation problem is formulated as:

$$\text{minimize } f(x) = x_1^2 + 6x_2^2 - 3x_1x_2 - x_1 + 2x_2$$

$$\text{s.t. } g_1(x) = x_1 - 12 \leq 0$$

$$g_2(x) = 7x_1 + 2x_2 - 4 \leq 0$$

$$g_3(x) = x_1 + x_2 - 2 \leq 0$$

The initial starting point for an optimization is  $x^0 = (0.5, 0)$  for which a search direction is determined to be  $d^0 = [-1, -0.5]$  and the value of Lagrange multipliers for the constraints is  $\lambda = 0$ .  
Let  $R_0 = 1$  and  $\gamma = 0.5$ . Choose the trial step according to the sequence  $t_0 = 1, t_1 = \frac{1}{2}, t_2 = \frac{1}{4}, t_3 = \frac{1}{8}$ .

Calculate the step size using the inexact line search procedure (approximate step size procedure) and determine the new design variables  $x^1$  for the next iteration.

Slide 8 - Lecture 7

Exercise 13.1

$$\Phi_0 = f_0 + R_0 V_0 \quad \leftarrow \text{current descent function} \quad \text{Slide 5}$$

$$f_0 = f(0.5, 0) = 0, \quad R_0 = 1, \quad V_0 = \max\{0, g_1, g_2, g_3\}$$

$$g_1(0.5, 0) = -11.5 < 0$$

$$g_2(0.5, 0) = -3.5 < 0$$

$$g_3(0.5, 0) = -1.5 < 0$$

$\uparrow$   
constraint violation function

Step 1: Form  $\Phi_0$

$$R_0 = \sum V_0 + \sum \lambda_i = 0 \quad \text{Sum of Lagrange Multipliers.}$$

$$\Phi_0 = \max\{R_0, f_0\} = 0$$

Slide 6

$$\text{Current descent function } \Phi = f_0 + R V_0 = 0 + 0 = 0$$

Slide 7

a step  $t = 1$  and  $\lambda = 0$

$$\begin{bmatrix} x^1 \\ x^2 \end{bmatrix} = \begin{bmatrix} x^0 \\ x^0 \end{bmatrix} + t \begin{bmatrix} d_1^0 \\ d_2^0 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} -1 \\ -0.5 \end{bmatrix} = \begin{bmatrix} -0.5 \\ -0.5 \end{bmatrix}$$

Cost and constraint functions at the new point

$$f(-0.5, -0.5) = -0.75$$

$$\left. \begin{aligned} g_1(-0.5, -0.5) &= -12.5 \\ g_2(-0.5, -0.5) &= -6.5 \end{aligned} \right\} \text{active } V_1 = \max\{0, -12.5, -6.5\} = -6.5$$

$$g_3(-0.5, -0.5) = -2$$



The descent function at the first trial point is calculated to

$$\Phi_{1,0} = f_{1,0} + P \cdot V_{1,0} = 0.75 + 1 \cdot 0 = 0.75$$

Trial 2

As  $\Phi_{1,0} > \Phi_0$ , the descent condition is violated and the process is repeated for a step size of  $t_2 = 0.5$  and  $j = 2$

$$\begin{bmatrix} x_1^{(1,1)} \\ x_2^{(1,1)} \end{bmatrix} = \begin{bmatrix} x_1^{(0)} \\ x_2^{(0)} \end{bmatrix} + t_1 \begin{bmatrix} d_1^{(0)} \\ d_2^{(0)} \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0 \end{bmatrix} + 0.5 \begin{bmatrix} -1 \\ -0.5 \end{bmatrix} = \begin{bmatrix} 0 \\ -0.25 \end{bmatrix}$$

Cost and constraint functions at the new point.

$$f(0, -0.25) = -0.125$$

$$g_1(0, -0.25) = -12$$

$$g_2(0, -0.25) = -3.5$$

$$g_3(0, -0.25) = -20.25$$

all inactive  $\Rightarrow V_{1,1} = 0$

Descent function at the second trial point is

$$\Phi_{1,1} = f_{1,1} + P \cdot V_{1,1} = -0.125 + 1 \cdot 0 = -0.125$$

$\Phi_{1,1} < \Phi_0 \Rightarrow$  not violated and we check if it fulfills the requirements.

$$\beta_1 = 8 \|d^{(0)}\|^2 = 0.5 \cdot ((-1)^2 + (-0.5)^2) = 0.625$$

$$\Phi_{1,1} + t_1 \beta_1 = -0.125 + 0.5 \cdot 0.625 = 0.1875 > \Phi_0 = 0 \Rightarrow \text{Violate}$$

Trial 3  $t_2 = 0.25$   $j = 2$

$$\begin{bmatrix} x_1^{(1,2)} \\ x_2^{(1,2)} \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0 \end{bmatrix} + 0.25 \begin{bmatrix} -1 \\ -0.5 \end{bmatrix} = \begin{bmatrix} 0.25 \\ -0.125 \end{bmatrix}$$

$$f(0.25, -0.125) = -0.1875$$

$$g_1(0.25, -0.125) = -11.75$$

$$g_2(0.25, -0.125) = -2$$

$$g_3(0.25, -0.125) = -19.875$$

Inactive  $\Rightarrow V_{1,2} = 0$

Descent function

$$\Phi_{1,2} = -0.1875 + 1 \cdot 0 = -0.1875 < \Phi_0 = 0 \Rightarrow \text{further check}$$

$$\Phi_{1,2} + t_2 \beta_1 = -0.1875 + 0.25 \cdot 0.625 = -0.03125 < \Phi_0$$

$\Rightarrow$  satisfies descent condition

Updated design is therefore  $\begin{bmatrix} x_1^{(1)} \\ x_2^{(1)} \end{bmatrix} = \begin{bmatrix} 0.25 \\ -0.125 \end{bmatrix}$

## Exercise 4

Solve the following problem by setting up the solution tree and using local minimization Branch & Bound method:

Minimize:  $f(x) = -6x_1 - 3x_2$

s.t.  $g_1(x) = 0.5x_1 + x_2 - 11 \leq 0$

$g_2(x) = 2.5x_1 + x_2 - 20 \leq 0$

$g_3(x) = 2x_1 - x_2 - 10 \leq 0$

Both  $x_1$  and  $x_2$  should be integer values.

Figure in exam can be used to help.

Use plot.m (And slides 16-18 in Lecture 7 and example)

Continuous guess of solution

$x_c^* = \begin{bmatrix} 4.5 \\ 8.5 \end{bmatrix} \Rightarrow f(x_c^*) = -52.5$

$x_1 \leq 4$

$x_c^* = \begin{bmatrix} 4 \\ \end{bmatrix} \Rightarrow f(x_c^*) = 51$

super  $x_2$  value at  $x_1 = 4$

$x_1 \geq 5$

$x_c^* = \begin{bmatrix} 5 \\ 7.5 \end{bmatrix} \Rightarrow f(x_c^*) =$

$x_2 \leq 7$

$x_2$  super  $x_1$  value at  $x_2 = 7$

discrete solution

$x^* = \begin{bmatrix} 5 \\ 7 \end{bmatrix} \Rightarrow f(x^*) = 51$

Two equally good solutions.



Solve the following linear optimisation problem using the basic steps of the simplex method and tableaux.

Minimize:  $f(x) = 3x_1 - 2x_2$

s.t.  $g_1(x) = -2x_1 + x_2 \leq 5$   
 $g_2(x) = x_1 + 4x_2 \leq 3$   
 $x_1 \geq 0$   
 $x_2 \geq 0$

(1) written as standard linear programming problem,

Minimize  $f(x) = 3x_1 - 2x_2$

s.t.  $g_1(x) = -2x_1 - x_2 + x_3 = 5$   
 $g_2(x) = x_1 + 4x_2 + x_4 = 3$  } convert inequalities to equalities.

$x_1 \geq 0$ ;  $x_2 \geq 0$ ;  $x_3 \geq 0$ ;  $x_4 \geq 0$

(2) or written in matrix form.

Minimize  $f(x) = c^T x$   
s.t.  $Ax = b$

$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$

consists of (objective function)

$c = \begin{bmatrix} 3 \\ -2 \\ 0 \\ 0 \end{bmatrix}$

consists of constraints

$A = \begin{bmatrix} -2 & 1 & 1 & 0 \\ 1 & 4 & 0 & 1 \end{bmatrix}$

right side of constraints  
 $b = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$

(3) Initial tableau

| Basic | $x_1$ | $x_2$ | $x_3$ | $x_4$ | b | ratio        |
|-------|-------|-------|-------|-------|---|--------------|
| $x_3$ | -2    | 1     | 1     | 0     | 5 | $5/1 = 5$    |
| $x_4$ | 1     | 4     | 0     | 1     | 3 | $3/4 = 0.75$ |
| Cost  | 3     | -2    | 0     | 0     | f |              |

negative cost (column first)

lower ratio

(4) basic and non basic variables are

basic:  $x_3 = 5$ ;  $x_4 = 3$  non-basic:  $x_1 = 0$ ;  $x_2 = 0$  Cost  $f = 0$

(5) Gauss elimination

| Basic | $x_1$ | $x_2$ | $x_3$ | $x_4$ | b         |
|-------|-------|-------|-------|-------|-----------|
| $x_3$ | -2.25 | 0     | 1     | -0.25 | 4.25      |
| $x_2$ | 0.25  | 1     | 0     | 0.25  | 0.75      |
| Cost  | 3.5   | 0     | 0     | 0.5   | $f + 1.5$ |

(1) Divide row 2 by 4

(2) subtract row 2 from row 1

(3) Add  $2 \times$  Row 2 to row 3

Basic:  $x_3 = 4.25$ ;  $x_2 = 0.75$  all positive  $\Rightarrow$  optimum.

Non-Basic:  $x_1 = 0$ ;  $x_4 = 0$

$f = -1.5$



The cost function  $f(x)$  is defined as the sum of squares of the residual function  $r(x)$

$$f(x) = \frac{1}{2} \sum_{i=1}^m r_i^2$$

Where

$$r(x) = \begin{bmatrix} x_1 - 2 \\ (x_1 - 2\beta) \cdot x_2 \\ x_2 + \end{bmatrix} \quad (5)$$

The initial cost function  $f(x_0) = 130$  for  $x_0 = [2, 7]$  and the constant  $\beta = 2$ . The minimum value  $f(x^*) \approx 0.396478$  at  $x^* \approx [2.0882, -0.2148]$ , see fig. 1.

1. Derive an analytical expression for Jacobian matrix associated with equation 5.

$$J = \begin{bmatrix} \frac{\partial r_1(x)}{\partial x_1} & \frac{\partial r_1(x)}{\partial x_2} \\ \frac{\partial r_2(x)}{\partial x_1} & \frac{\partial r_2(x)}{\partial x_2} \\ \frac{\partial r_3(x)}{\partial x_1} & \frac{\partial r_3(x)}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ x_2 & x_1 - 2\beta \\ 0 & 1 \end{bmatrix}$$

2. Numerical solutions are expected for the following two sub-questions where the initial parameters  $x_0 = [2, 7]$  are inserted into the expression derived above.

a) Using the initial parameters  $x_0 = [2, 7]$ ; Calculate the gradient and approximate the Hessian using only first order information from the Jacobian matrix derived above in question 1.

• Hessian based on first order information

$$H = J^T J = \begin{bmatrix} 1 & 7 & 0 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 7 & -2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 50 & -14 \\ -14 & 5 \end{bmatrix}$$

• Gradient based on first order information

$$\nabla f(x) = J^T \cdot r = \begin{bmatrix} 1 & 7 & 0 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ -14 \\ 8 \end{bmatrix} = \begin{bmatrix} -98 \\ 36 \end{bmatrix}$$

5) Complete the first 3 iterations using the Levenberg-Marquardt algorithm (where the step was defined as  $S_k = -(H_k + \lambda I)^{-1} \nabla f(x_k)$ ). Start the iterations using an initial damping parameter  $\lambda = 10$  and report  $\lambda$ , object function  $f(x_k)$  and  $x_k$  for the first 3 iterations.

Using Texas

$$k=0 \quad x_{k+1} = x_k - (H_k + \lambda I)^{-1} \nabla f(x_k) = x_k + S_k$$

$$x_1 = \begin{bmatrix} 2 \\ 7 \end{bmatrix} - \left( \begin{bmatrix} 50 & -14 \\ -14 & 5 \end{bmatrix} + 10 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right)^{-1} \cdot \begin{bmatrix} -98 \\ -36 \end{bmatrix} = \begin{bmatrix} 3.37 \\ 5.88 \end{bmatrix}$$

$$f(x_1) = \frac{1}{2} \sum r_i^2 = 0.5 \cdot (1.37^2 + 3.88^2 + 6.88^2) = 31.13 \quad \text{Use } x_1 \text{ to get } r_i$$

$$\text{if } f(x_k) < f(x_{k+1}) \Rightarrow \lambda = \frac{1}{2} \lambda$$

$$31.13 < 130 \Rightarrow \lambda = \frac{1}{2} \cdot 10 = 5$$

$$k = k + 1 = 0 + 1 = 1$$

Update  $\nabla f(x_k)$  and  $H_k$

$$x_2 = \begin{bmatrix} 3.37 \\ 5.88 \end{bmatrix} - \left( \begin{bmatrix} 35.57 & -3.7 \\ -3.7 & 1.392 \end{bmatrix} + 5 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right)^{-1} \cdot \begin{bmatrix} -20.41 \\ 9.214 \end{bmatrix} = \begin{bmatrix} 3.76 \\ 4.67 \end{bmatrix}$$

$$f(x_2) = 0.5 \cdot \sum r_i^2 = 0.5 \cdot (1.76^2 + 1.12^2 + 5.17^2) = 18.25$$

$$f(x_k) < f(x_{k+1}) \Rightarrow \lambda = \frac{1}{2} \lambda$$

$$18.25 < 31.13 \Rightarrow \lambda = \frac{1}{2} \cdot 5 = 2.5$$

$$k = k + 1 = 1 + 1 = 2$$

continue

solution lecture 10 Ex 10

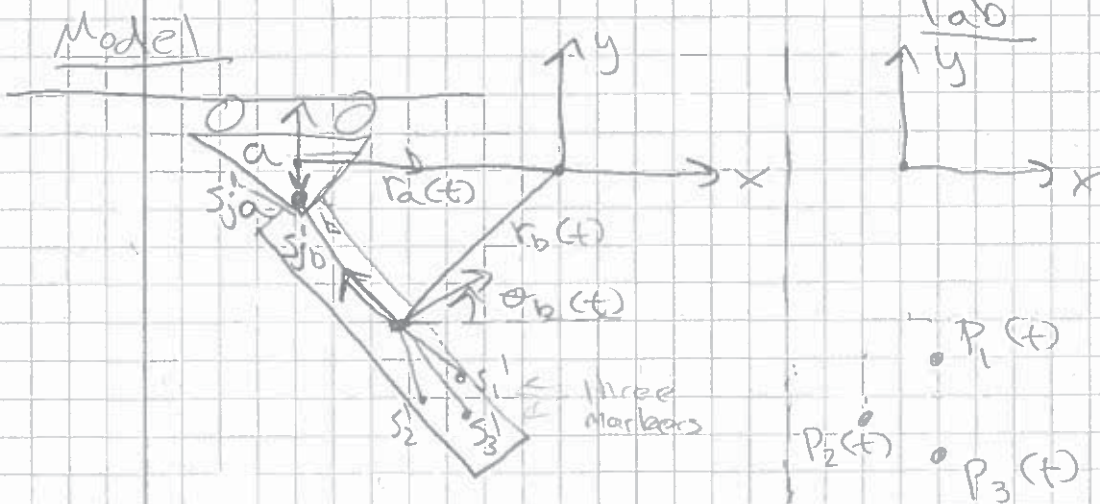
↓ Solve Exam 10

4. Use 21-23 Lecture

|     | Texas        |       |           | Matlab       |       |           |
|-----|--------------|-------|-----------|--------------|-------|-----------|
| ite | x            | f(x)  | $\lambda$ | x            | f(x)  | $\lambda$ |
| 0   | [2, 7]       | 130   | 10        | [2, 7]       | 130   | 10        |
| 1   | [3.3, 5.88]  | 31.13 | 5         | [3.3, 5.88]  | 31.43 | 5         |
| 2   | [3.76, 4.67] | 18.25 | 2.5       | [3.76, 4.67] | 18.23 | 2.5       |
| 3   |              |       |           |              |       |           |



# Exercise 1



2 rigid bodies: Translation to ground, revolute in-between

$$r_a(t) = \begin{bmatrix} x_a(t) \\ y_a(t) \end{bmatrix}, \quad r_b(t) = \begin{bmatrix} x_b(t) \\ y_b(t) \end{bmatrix}, \quad \theta_a(t), \theta_b(t)$$

1/ Is the system kinematically under-determined, determinate or over-determined?  
 - describe position and rotation of the bodies

DOF:  $M - \text{Links} - 2 \cdot \text{joints} - 3 \cdot \text{grounded}$   
 $1 = 3 - 2 \cdot 2 - 3 \cdot 1 = 2 \text{ DOF}$   
 2. bodies + ground

The number of drivers: As the system is driven by 3 measured markers, it means

$$n \text{ DOF} < n \text{ Drivers} \Rightarrow \text{Over-Determined}$$

2/ Formulate a constrained non-linear least-squares optimisation, that can be used to compute position  $r_a(t)$ ,  $r_b(t)$  and rotations  $\theta_a$  and  $\theta_b$

Translation:  $r_{aj}$   
 Revolute:  $r_a + A(\theta_a) \cdot s_{ja} - (r_b + A(\theta_b) \cdot s_{jb}) = 0$   
 Constraint:  $\theta_A$

$$r(r_a, r_b, \theta_a, \theta_b) = \begin{bmatrix} r_b + A_b \cdot s_1 - p_1 \\ r_b + A_b \cdot s_2 - p_2 \\ r_b + A_b \cdot s_3 - p_3 \end{bmatrix} \quad \text{and} \quad f = \frac{1}{2} \sum \|r_i\|^2$$

(f = f(r\_a, r\_b, \theta\_a, \theta\_b))

1/ minimize  $f = \frac{1}{2} \sum \|r_i\|^2$

st  $\Phi = 0$

$\Phi(r_a, r_b, \theta_a, \theta_b)$

Probably need to go  $A \rightarrow B$  way to obtain enough variables.



3) Derive the first order optimality conditions for the optimisation problem derived in question 2.

The first order order optimality condition is checked by:

$$\mathcal{L} = f(x) + \sum V_i h_i(x) + \sum U_j g_j(x) + \frac{1}{2} s_i^2 \quad (\text{Lagrangian})$$

skipped notation:  $Gq + \frac{1}{2} \Phi \leq \lambda = 0$

$$\mathcal{L} = \frac{1}{2} \left( (r_b + A_b \cdot s_1' - p_1)^2 + (r_b + A_b \cdot s_2' - p_2)^2 + (r_b + A_b \cdot s_3' - p_3)^2 \right) \\ + V_1 (r_a) + V_2 (r_a + A_a \cdot (-s_{ja}') - (r_b + A_b \cdot s_{jb}')) + V_3 (\Theta_A)$$

vec 2  
slide 18

and  $\frac{\partial \mathcal{L}}{\partial x_k} = 0$   $\int \frac{\partial \mathcal{L}}{\partial v_j} = 0$  and we know  $\nabla h = \text{Jacob}$

## Exercise 8

Pipe pressures, 10 pipes  $\rightarrow$  constraints are minimum cross sectional values and maximum.

a) Explain the main mathematical difficulty of solving the optimization problem of minimizing the maximum pressure in the  $n$  pipes.

- As it is a min-max problem, i.e. multi-objective function the main difficulty is to form a scalar objective as a combination of multiple objectives.  
(as, non-differentiability for gradient do not seem a problem)

b) State the mathematical formulation for three fundamentally different ways of formulating the optimization problem. Describe advantages and disadvantages of the three chosen optimization problem formulations.

■ Weighted sum:  $f_{\text{wsum}}(x) = f_{\text{wsum}}(f_k(x)) = \sum_{k=1}^{n_0} w_k f_k$

- global criterion  
Pick up one  
Point as fo
- Advantage: Simple, the pressures in each pipe is weighted appropriately also pipe important - Pressures
  - Disadvantage: Crude, choose weight factors

■ Max-min:  $f(x) = \max_{k=1, \dots, n_0} f_k(x)$

- Advantage: Simple, as only max value is taken, fast. The max pressure is used, not something scaled.
- Disadvantage: Crude, only looking at max-pressure pipe! Too crude

■ P-norm:  $f_p(x) = \left( \sum_{k=1}^{n_0} f_k^p \right)^{1/p}$

- Advantage: Largest values are weighted most
- Disadvantage: P-parameter should be chosen with care